

D830

Introduction to Cryptography

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A video streaming company with eight data centers must encrypt 40 TB of stored video quickly. Its keys live in a central key management system. Which approach fits?

- A. Hash the video with SHA-256.
- B. Use asymmetric encryption for everything, since it is always more secure.
- C. Encrypt the video with a symmetric cipher such as AES.
- D. Encrypt every file directly with RSA.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A consulting firm with offices in four cities has 50 employees, and each must be able to exchange private messages with every other employee. Using only symmetric keys, one unique key per pair, how many keys are needed?

- A. 2,500
- B. 100
- C. 2,450
- D. 1,225

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

The same 50 employees switch to asymmetric encryption, each with a public/private key pair. How many keys exist in total?

- A. 1,225
- B. 2,500
- C. 50
- D. 100

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An engineer at a medical device maker wants to send a design file that only a partner company can read. The partner's certificate was issued last year by a well-known certificate authority. Which key should encrypt the file (or the file's session key)?

- A. The partner's public key
- B. The engineer's public key
- C. The engineer's private key
- D. The partner's private key

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Your system receives a signed software update from a vendor. Which key verifies the signature?

- A. Your own private key
- B. A shared symmetric key
- C. The vendor's public key
- D. The vendor's private key

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A web app with 250,000 users, hosted in a low-latency cloud region, is redesigning how it stores user passwords. What is the best practice?

- A. Store plain text but restrict database access.
- B. Store a unique salt and a slow password hash such as Argon2, bcrypt, or PBKDF2.
- C. Store an unsalted SHA-256 hash.
- D. Encrypt passwords with AES and store the key on the server.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A law firm with 90 partners digitally signs the contracts it sends by email. What does the signature alone provide?

- A. Integrity, proof of origin, and non-repudiation
- B. Integrity, proof of origin, non-repudiation, and confidentiality
- C. Confidentiality, so only the recipient can read it
- D. Integrity only, with no link to the signer

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A contract must be kept secret from everyone but the recipient, and the recipient must be able to prove who sent it. What is needed?

- A.** Send a hash of the contract.
- B.** Sign with the sender's private key and encrypt with the recipient's public key.
- C.** Sign it with the sender's private key only.
- D.** Encrypt it with the sender's private key only.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

An imaging company encrypts a bitmap logo with AES in ECB mode. The key is 256 bits and stored in a hardware security module, yet the encrypted image still shows the logo's outline. What is the cause and fix?

- A.** The key is too short; use a longer key.
- B.** Move the key out of the hardware security module.
- C.** ECB encrypts identical blocks identically; switch to an authenticated mode such as GCM with unique nonces.
- D.** ECB is safe for all data except images.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Staff at a 2,000-person insurer get browser warnings on the internal HR site. Its certificate was issued by the company's own internal CA, whose root certificate is not in the laptops' trust stores. The site uses TLS 1.3. What is the best fix?

- A.** Tell staff to click past the warning.
- B.** Deploy the internal root CA certificate to managed laptops' trust stores.
- C.** Turn off TLS on the HR site.
- D.** Upgrade to a newer TLS version.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A records agency with 16 departments still runs an archive system from 2003 that uses a 56-bit key. If the key were lengthened to 64 bits, how many times larger would the key space be?

- A.** 8 times
- B.** 2^{64} times
- C.** About 1.14 times
- D.** 256 times

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12. [TRANSFER — reversed stem / different variable]

A key space must grow by a factor of 1,024. How many bits must be added to the key?

- A.** 10
- B.** 1,024
- C.** 5, since each bit multiplies the space by 4
- D.** 512

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A video streaming company ~~with eight data centers~~ must encrypt 40 TB of stored video quickly. ~~Its keys live in a central key management system.~~ Which approach fits?

Target word: cipher type for bulk data **Decides:** Large volume that must be encrypted fast.

A	Confusion set	Hashes cannot be reversed, so the video could never be played.
B	Absolute language	Security depends on proper use and key size, not on the family.
C	Correct	Symmetric ciphers are built for fast bulk encryption.
D	Adjacent category	RSA is far too slow and size-limited for bulk data; it protects keys.

2. Answer D Concept 2

A consulting firm ~~with offices in four cities~~ has 50 employees, and each must be able to exchange private messages with every other employee. Using only symmetric keys, one unique key per pair, how many keys are needed?

Target word: number of pairwise keys **Decides:** $n(n - 1) \div 2$ with $n = 50$.

A	Inverted operation	50^2 counts each person pairing with themselves and each pair twice.
B	Confusion set	That is the asymmetric count, $2n$.
C	Dropped element	50×49 forgets that each key serves two people.
D	Correct	$50 \times 49 \div 2 = 1,225$.

3. Answer D Concept 2 · transfer

The same 50 employees switch to asymmetric encryption, each with a public/private key pair. How many keys exist in total?

Target word: asymmetric key count **Decides:** 2 keys per user.

A	Confusion set	The symmetric pairwise count.
B	Inverted operation	Squares the users instead of doubling.
C	Dropped element	Counts only one key per user.
D	Correct	$50 \text{ users} \times 2 \text{ keys}$.

4. Answer A Concept 3

An engineer at a medical device maker wants to send a design file that only a partner company can read. The partner's certificate ~~was issued last year by a well-known certificate authority.~~ Which key should encrypt the file (or the file's session key)?

Target word: key for confidentiality **Decides:** Only the partner must be able to decrypt.

A	Correct	Only the partner's private key can then decrypt it.
B	Confusion set	Only the engineer could decrypt the result.
C	Inverted operation	That creates a signature anyone can check, not secrecy.
D	Confusion set	The engineer never has it; only the partner holds it.

5. Answer C Concept 3 · transfer

Your system receives a signed software update from a vendor. Which key verifies the signature?

Target word: reverse: key for verification **Decides:** Signed with the vendor's private key.

A	Confusion set	Your keys play no part in checking the vendor's signature.
B	Adjacent category	That would be a MAC, not a digital signature.
C	Correct	The public half checks what the private half signed.
D	Inverted operation	Used to create the signature, and never shared.

6. Answer B Concept 4

A web app with 250,000 users, ~~hosted in a low-latency cloud region~~, is redesigning how it stores user passwords. What is the best practice?

Target word: password storage method **Decides:** Passwords should never be recoverable.

A	Adjacent category	Access control alone fails if the database is ever exposed.
B	Correct	Salting defeats precomputed tables; slowness limits guessing.
C	Dropped element	Drops the salt and the slowness; fast unsalted hashes are easy to guess at scale.
D	Confusion set	Encryption is reversible; anyone with the key recovers every password.

7. Answer A Concept 5

A law firm ~~with 90 partners~~ digitally signs the contracts it sends by email. What does the signature alone provide?

Target word: properties of a signature **Decides:** Signature, no encryption.

A	Correct	Signatures do not hide content.
B	Absolute language	Adds confidentiality, which a signature never gives.
C	Confusion set	Signed content is still readable by anyone.
D	Dropped element	Drops authentication and non-repudiation.

8. Answer B Concept 5 · transfer

A contract must be kept secret from everyone but the recipient, and the recipient must be able to prove who sent it. What is needed?

Target word: reverse: from requirements to operations **Decides:** Secrecy and proof of origin are separate operations.

A	Adjacent category	A bare hash gives neither secrecy nor proof of sender.
B	Correct	Signature gives proof of origin; encryption gives secrecy.
C	Dropped element	No confidentiality.
D	Inverted operation	Anyone with the sender's public key can read it.

9. Answer C Concept 6

An imaging company encrypts a bitmap logo with AES in ECB mode. ~~The key is 256 bits and stored in a hardware security module,~~ yet the encrypted image still shows the logo's outline. What is the cause and fix?

Target word: cause of pattern leakage **Decides:** ECB mode.

A	Adjacent category	Key length does not change ECB's pattern leak.
B	Camouflage	Key storage does not decide the leak and would weaken protection.
C	Correct	Chained or counter modes hide repeated patterns.
D	Absolute language	ECB leaks patterns in any structured data.

10. Answer B Concept 7

Staff at a ~~2,000-person~~ insurer get browser warnings on the internal HR site. Its certificate was issued by the company's own internal CA, whose root certificate is not in the laptops' trust stores. ~~The site uses TLS 1.3.~~ What is the best fix?

Target word: fix a certificate trust error **Decides:** The issuing CA is not trusted by the devices.

A	Dropped element	Drops trust validation and trains users to ignore real warnings.
B	Correct	Once the CA is trusted, the chain validates.
C	Inverted operation	Removes encryption instead of fixing trust.
D	Camouflage	Protocol version does not decide certificate trust.

11. Answer D Concept 8

A records agency ~~with 16 departments~~ still runs an archive system ~~from 2003~~ that uses a 56-bit key. If the key were lengthened to 64 bits, how many times larger would the key space be?

Target word: growth in key space **Decides:** 8 added bits = 2^8 .

A	Dropped element	Counts bits, not doublings.
B	Wrong stem number	That is the whole 64-bit key space, not the increase.
C	Adjacent category	$64 \div 56$ treats bits as linear.
D	Correct	$2^8 = 256$.

12. Answer A Concept 8 · transfer

A key space must grow by a factor of 1,024. How many bits must be added to the key?

Target word: solve for added bits **Decides:** $2^k = 1,024$.

A	Correct	$2^{10} = 1,024$.
B	Wrong stem number	The growth factor, not the bits.
C	Confusion set	Each bit doubles the space.
D	Inverted operation	Halves the factor instead of taking log base 2.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Symmetric vs. asymmetric roles	Symmetric ciphers (e.g., AES) encrypt bulk data fast; asymmetric ciphers solve key exchange and signatures, so real systems use both (hybrid).	Encrypting large data directly with RSA, or thinking asymmetric is always “stronger.”	Large volumes of data vs. two parties who have never shared a key.
2	Key counts	Pairwise symmetric keys for n users = $n(n - 1) \div 2$; asymmetric needs $2n$ keys (one pair per user).	Using n^2 or forgetting to divide by 2.	A number of users who all need private channels.
3	Which key does what	Encrypt with the recipient’s public key; sign with the sender’s private key; verify with the sender’s public key.	Encrypting with your own private key to keep something secret.	“Only the recipient can read” vs. “prove who sent it.”
4	Hashing vs. encryption	Hashes are one-way and fixed-length; passwords are stored as salted, deliberately slow hashes, never as reversible ciphertext.	Encrypting passwords or using a fast, unsalted hash.	Password storage, or data that must never be recovered.
5	What digital signatures provide	Signatures give integrity, origin authentication, and non-repudiation — not confidentiality.	Believing a signed message is secret.	A signed document and a question about secrecy.
6	Modes of operation	ECB encrypts identical blocks identically and leaks patterns; use an authenticated mode such as GCM with a nonce that never repeats under the same key.	Blaming key length for pattern leakage.	Encrypted output that still shows structure.
7	PKI and certificate trust	A certificate is trusted when a CA in the device’s trust store vouches for it; fix trust errors by fixing the chain, not by clicking past warnings.	Training users to ignore certificate warnings.	Browser warnings on an internal or new site.
8	Key length and key space	Each added key bit doubles the key space: adding k bits multiplies it by 2^k .	Treating added bits as a linear increase.	Two key lengths in bits.

Course scope (public description): Introduction to Cryptography introduces students to foundational cryptography concepts and explore implementing encryption methods with symmetric and asymmetric algorithms. Students analyze principles and operations of cryptographic algorithms and protocols in secure systems. Students use cryptography frameworks and standards in alignment with organizational and information security guidelines.